SIMATS ENGINEERING

ASSESSMENT TOOL 3

FUNDAMENTALS OF DATA SCIENCE

Course Code: DSA04

CO3: Descriptive Statistics, Outlier Detection, Distribution Analysis and Statistical Inference

CASE STUDY — ANSWERS

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Code of Conduct

I, MOTHUKURI MARUTHI (Reg. No. 192425257), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Note: The supplied assessment describes scenarios but does not provide numerical datasets for these 20 case studies. Therefore, the answers explain the correct statistical methods and testing procedures without inventing numerical results. Where actual data are collected, calculate the stated statistics and report the resulting values.

General statistical approach

- Descriptive statistics: mean, median, standard deviation, quartiles, minimum, maximum, and interquartile range (IQR). Use histograms and boxplots to examine distribution and outliers.
- Outlier screening: IQR fences are $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$. A z-score (often $|z| > 3$) may also flag unusual values when its assumptions are reasonable.
- Do not automatically delete outliers. Check for data-entry or measurement errors first; valid rare events may be important. Consider robust summaries, transformations, or sensitivity analysis.
- Hypothesis tests: define H_0 and H_1, choose α (commonly 0.05), check assumptions, calculate a test statistic and p-value, then state the decision in context.
- A p-value below α supports rejecting H_0; it does not measure the size or practical importance of an effect. Report confidence intervals and effect sizes where possible.

CASE STUDY 1: E-Commerce Customer Analysis

Scenario

Customer transactions include purchase amount, number of orders, browsing time, and ratings. The aim is to understand purchasing behaviour and improve marketing.

A. Descriptive statistics

Calculate the mean and median purchase amount to describe typical spending; the standard deviation describes variability. Report Q1, Q2 (median), and Q3, plus the IQR = Q3 − Q1, to show the middle 50% of spending. Summarize orders, browsing time, and ratings similarly. Use histograms and boxplots; a right-skewed purchase distribution may have a mean above its median.

B. Outliers

Very high purchases may be bulk orders or premium customers; very low values may be refunds, test transactions, or genuine small purchases. Flag values outside Q1 − 1.5×IQR and Q3 + 1.5×IQR, inspect boxplots, and check transaction records. Correct confirmed errors; retain valid unusual purchases and consider median/IQR or a log scale for skewed spending.

C. Premium versus regular members

Let μP and μR be mean purchase amounts. H_0: $\mu P = \mu R$; H_1: $\mu P > \mu R$. Separate customers by membership, inspect distributions and variance, then use an independent two-sample t-test (Welch's test if variances differ). If assumptions are poor, use a suitable robust or Mann–Whitney analysis. At $\alpha = 0.05$, reject H_0 when the one-sided p-value is below 0.05; report the mean difference and confidence interval. Avoid claiming membership causes spending unless the study design supports causality.

Conclusion

Use the descriptive summaries to understand typical values and variation, investigate unusual observations before deciding how to treat them, and use a correctly specified hypothesis test with a confidence interval to support decisions. Interpret statistical results alongside practical context and data quality.

CASE STUDY 2: Renewable Energy Production Analysis

Scenario

Daily energy generation, sunlight hours, panel temperature, and efficiency are recorded across solar plants. Management wants to assess performance against a target.

A. Descriptive statistics and distribution

For each plant and overall, calculate mean, median, standard deviation, quartiles, minimum and maximum daily generation. Plot histograms, boxplots, and time-series charts. Compare mean generation with the expected target, and examine distribution shape, seasonal patterns, and variation between plants. Explore relationships with sunlight hours, panel temperature, and efficiency using scatter plots and correlations.

B. Abnormal generation and outliers

Unusually low values can reflect cloudy weather, equipment faults, or downtime; unusually high values may be weather-related or recording errors. Flag outliers with IQR fences or robust z-scores, then compare them with weather, maintenance, and sensor logs. Correct verified data errors, mark genuine outages or extreme weather, and avoid deleting valid events. Use robust summaries or analyse normal-operation and outage periods separately.

C. Confidence interval and hypothesis test

Let μ be mean daily generation and T the expected target. H_0: $\mu = T$; H_1: $\mu \neq T$. Calculate a confidence interval for μ and perform a one-sample t-test if observations are reasonably independent and assumptions are adequate. If $p < 0.05$, reject H_0 and conclude the mean differs from target; otherwise, evidence is insufficient to show a difference. Report the mean, interval, p-value, and practical size of any shortfall or surplus.

Conclusion

Use the descriptive summaries to understand typical values and variation, investigate unusual observations before deciding how to treat them, and use a correctly specified hypothesis test with a confidence interval to support decisions. Interpret statistical results alongside practical context and data quality.

CASE STUDY 3: Hospital Patient Recovery Analysis

Scenario

Patient data include age, treatment cost, hospital days, recovery time, and satisfaction score. The hospital wants to evaluate treatment effectiveness and care quality.

A. Descriptive statistics

For recovery time and treatment cost, report mean, median, standard deviation, quartiles, IQR, minimum and maximum. Recovery time and cost often have right-skewed distributions, so the median and IQR can better represent typical patients. Use histograms and boxplots; stratify results by treatment type and relevant patient characteristics.

B. Outliers

Long stays and high costs may be caused by severe illness, complications, or unusual care needs; they may also arise from data-entry mistakes. Use boxplots, IQR fences, and record-level checks. Verify extreme records with clinical and billing information. Do not remove valid severe cases; report robust summaries and consider separate analysis by diagnosis or severity.

C. New treatment versus existing treatment

Let μN and μE be mean recovery times. H_0: $\mu N = \mu E$; H_1: $\mu N < \mu E$. Compare independent treatment groups using Welch's two-sample t-test, or use a paired test if the design truly matches patients. Random assignment or adjustment for baseline severity is important. At $\alpha = 0.05$, a one-sided $p < 0.05$ supports faster average recovery with the new treatment; report the estimated time difference and confidence interval.

Conclusion

Use the descriptive summaries to understand typical values and variation, investigate unusual observations before deciding how to treat them, and use a correctly specified hypothesis test with a confidence interval to support decisions. Interpret statistical results alongside practical context and data quality.

CASE STUDY 4: Student Academic Performance Analysis

Scenario

Data include attendance, assignment marks, internal scores, study hours, and final exam marks.

A. Descriptive statistics

Calculate mean, median, standard deviation, quartiles, and range for marks and attendance. Histograms show distribution and possible ceiling effects; boxplots compare classes or attendance groups. Scatter plots can show how attendance, study hours, and assessment marks relate to final marks, without assuming that correlation proves causation.

B. Extreme marks

Very low marks may indicate learning difficulties, absence, or data errors; very high marks may be genuine exceptional performance or recording mistakes. Detect using boxplots, IQR fences, and checks against original scripts or grade records. Correct errors, retain valid scores, and report results both with and without influential observations if appropriate.

C. Attendance above versus below 75%

Let μH and μL be mean final marks for attendance >75% and <75%. H_0: $\mu H = \mu L$; H_1: $\mu H > \mu L$. Split students according to the stated threshold, inspect group sizes and distributions, and use Welch's independent t-test. If assumptions are unsuitable, use a robust/non-parametric alternative. Report p-value, mean difference, and CI; acknowledge confounding factors such as prior achievement and study time.

Conclusion

Use the descriptive summaries to understand typical values and variation, investigate unusual observations before deciding how to treat them, and use a correctly specified hypothesis test with a confidence interval to support decisions. Interpret statistical results alongside practical context and data quality.

CASE STUDY 5: Banking Loan Approval Analysis

Scenario

Loan application data include income, credit score, loan amount, repayment history, employment status, and approval status.

A. Descriptive statistics

Summarize income, credit score, and loan amount with mean, median, standard deviation, quartiles, range, and missing-value counts. Income and loan amounts may be skewed, so medians and IQRs are useful. Compare summaries by approval status and employment group, and visualize distributions with histograms and boxplots.

B. Outliers

Very high loan amounts or low credit scores can strongly influence averages and risk models. Use IQR fences, robust z-scores, scatter plots, and validation against original application records. Check whether the value is an error, a legitimate high-risk application, or a rare but valid customer. Do not automatically exclude genuine applications; use robust modelling and sensitivity checks.

C. Credit score and approval

Define approval as a binary outcome. H_0: approval probability is equal for higher- and lower-credit-score groups; H_1: the higher-score group has a greater approval probability. Compare proportions with a two-proportion z-test or chi-square test, or fit logistic regression with credit score and relevant covariates. Report the effect size, CI, and p-value. Distinguish statistical association from fairness and causal claims; audit for biased decision patterns.

Conclusion

Use the descriptive summaries to understand typical values and variation, investigate unusual observations before deciding how to treat them, and use a correctly specified hypothesis test with a confidence interval to support decisions. Interpret statistical results alongside practical context and data quality.

CASE STUDY 6: Online Learning Platform Analysis

Scenario

Learner data include completion percentage, quiz scores, video watch time, login frequency, and feedback ratings.

A. Descriptive statistics

Use mean, median, standard deviation, quartiles, and range for quiz scores, watch time, and completion rate. Completion percentages are bounded between 0 and 100 and may cluster near 0 or 100. Use histograms, boxplots, and scatter plots to compare engagement measures with quiz scores.

B. Activity outliers

Very high watch time may reflect repeated viewing or a session left open; very low activity may reflect disengagement or tracking failure. Flag with IQR fences and inspect event logs, timestamps, and device/session behaviour. Correct measurement errors, retain genuine activity, and consider robust summaries or capped/log-transformed values only when justified.

C. Watch more than 70% versus others

Let μH and μL be mean quiz scores for learners watching >70% and the remaining learners. H_0: $\mu H = \mu L$; H_1: $\mu H > \mu L$. Compare groups using Welch's t-test or a suitable non-parametric test. Inspect distributions and sample sizes, report effect size and CI, and consider course difficulty and prior ability as confounders. Association alone does not prove that watching more causes higher scores.

Conclusion

Use the descriptive summaries to understand typical values and variation, investigate unusual observations before deciding how to treat them, and use a correctly specified hypothesis test with a confidence interval to support decisions. Interpret statistical results alongside practical context and data quality.

CASE STUDY 7: Retail Sales Performance Analysis

Scenario

Branch data include daily sales, customer count, discount percentage, product category, and customer feedback.

A. Descriptive statistics

For daily sales and customer counts, calculate mean, median, standard deviation, quartiles, IQR, minimum, and maximum. Plot histograms and time-series charts. Summarize by branch, weekday, product category, and season to distinguish ordinary performance from recurring patterns.

B. Abnormal sales

Festival sales may be genuine seasonal peaks; very low sales may reflect closures, stockouts, or system failures. Detect unusual values with boxplots, IQR fences, robust z-scores, and time-series residuals relative to weekday/season patterns. Verify promotions, store hours, and transaction systems. Retain valid festival days but analyse them separately when estimating typical sales.

C. Urban versus rural branches

Let μU and μR be mean sales. H_0: $\mu U = \mu R$; H_1: $\mu U > \mu R$. Calculate each group mean and a confidence interval for the difference, then use Welch's two-sample t-test. If branches are repeatedly observed, use a model accounting for branch and time. Reject H_0 if the one-sided p-value is below α; interpret the difference in business terms and account for store size and customer traffic.

Conclusion

Use the descriptive summaries to understand typical values and variation, investigate unusual observations before deciding how to treat them, and use a correctly specified hypothesis test with a confidence interval to support decisions. Interpret statistical results alongside practical context and data quality.

CASE STUDY 8: Food Delivery Service Analysis

Scenario

Order data include delivery time, order value, rating, travel distance, and delivery partner experience.

A. Descriptive statistics

Calculate mean, median, standard deviation, quartiles, IQR, and range for delivery time and customer ratings. The median and upper quartiles help describe typical and slower deliveries. Use histograms, boxplots, and scatter plots against distance and partner experience.

B. Long delivery times

Long times can reduce satisfaction, but may be caused by long distances, weather, restaurant preparation, or unusual traffic. Flag outliers with IQR fences and compare timestamps with order records. Correct timestamp errors, retain genuine delays, and report median and 90th/95th percentile times. Analyse by distance, restaurant, and peak period.

C. Experienced versus new partners

Let μE and μN be mean delivery times. H_0: $\mu E = \mu N$; H_1: $\mu E < \mu N$. Compare groups using Welch's t-test, or a regression adjusting for distance, traffic, and order type. A one-sided $p < 0.05$ supports shorter mean delivery time among experienced partners. Report the time difference and CI; ensure groups are comparable.

Conclusion

Use the descriptive summaries to understand typical values and variation, investigate unusual observations before deciding how to treat them, and use a correctly specified hypothesis test with a confidence interval to support decisions. Interpret statistical results alongside practical context and data quality.

CASE STUDY 9: Manufacturing Defect Analysis

Scenario

Production data include units produced, defect count, machine temperature, production time, and operator shift.

A. Descriptive statistics

Summarize defect counts and production time with means, medians, SDs, quartiles, and ranges; summarize machine temperature similarly. Also calculate defect rate = defective units / total units. Use run charts and histograms to detect shifts, and compare rates by machine, product, and shift.

B. Abnormal defect values

A sudden high defect count may signal machine failure, material problems, or a genuine process change. Detect unusual values using control charts, IQR fences, and time-series checks. Verify production volume and inspection records; counts should be interpreted relative to units produced. Correct data errors but preserve real defect spikes because they are essential quality signals.

C. Day versus night defect rate

Let pD and pN be defect proportions. H_0: pD = pN; H_1: pD ≠ pN. Use a two-proportion test or chi-square test based on defective and inspected units for each shift. If counts are small, use Fisher's exact test. Report the difference in rates, CI, and p-value, while accounting for machine, product mix, and production volume.

Conclusion

Use the descriptive summaries to understand typical values and variation, investigate unusual observations before deciding how to treat them, and use a correctly specified hypothesis test with a confidence interval to support decisions. Interpret statistical results alongside practical context and data quality.

CASE STUDY 10: Mobile Network Performance Analysis

Scenario

Network data include signal strength, internet speed, call drop rate, user count, and complaints by location.

A. Descriptive statistics

Report mean, median, SD, quartiles, and range for internet speed, signal strength, and call-drop rate. Use histograms and boxplots by location and time. Examine distributions separately for urban and rural areas and use scatter plots to explore relationships with user count and complaints.

B. Extreme values

Very low speed or high call-drop values may be real outages or measurement problems. Detect them with IQR fences, robust z-scores, time-series monitoring, and device/server logs. Verify timestamp, device, and location records. Retain genuine outages, flag them for incident analysis, and avoid treating real service failures as disposable noise.

C. Urban versus rural speed

Let μU and μR be mean internet speeds. H_0: $\mu U = \mu R$; H_1: $\mu U \neq \mu R$. Construct confidence intervals and use Welch's t-test, or a regression that accounts for repeated measurements and time of day. Check sample comparability and skewness. Report mean difference, CI, p-value, and practical impact on users.

Conclusion

Use the descriptive summaries to understand typical values and variation, investigate unusual observations before deciding how to treat them, and use a correctly specified hypothesis test with a confidence interval to support decisions. Interpret statistical results alongside practical context and data quality.

CASE STUDY 11: Public Transportation Usage Analysis

Scenario

Bus data include passenger count, ticket revenue, waiting time, travel duration, and route distance.

A. Descriptive statistics

Calculate mean, median, SD, quartiles, IQR, and range for passenger count, waiting time, and travel duration. Use route-wise summaries and time-series plots by hour/day. Median and upper percentiles can describe passenger waiting experience better than the mean alone.

B. Outliers

High passenger counts during peak hours may be expected, while very long waits may indicate congestion, missed service, or data error. Use boxplots and IQR fences within comparable route/time groups, plus control charts for waiting times. Verify schedules and logs. Retain valid peak values and use them to plan capacity.

C. Peak versus non-peak waiting time

Let μP and μN be mean waiting times. H_0: $\mu P = \mu N$; H_1: $\mu P > \mu N$. Compare independent observations using Welch's t-test, or paired comparisons by route/day where appropriate. Report mean difference, CI, and p-value. If $p < 0.05$, evidence supports longer peak waiting times; recommend schedule changes based on effect size.

Conclusion

Use the descriptive summaries to understand typical values and variation, investigate unusual observations before deciding how to treat them, and use a correctly specified hypothesis test with a confidence interval to support decisions. Interpret statistical results alongside practical context and data quality.

CASE STUDY 12: Agriculture Crop Yield Analysis

Scenario

Farm data include rainfall, fertilizer use, soil moisture, temperature, and crop yield.

A. Descriptive statistics

Summarize crop yield, rainfall, and fertilizer use with mean, median, SD, quartiles, and range. Use histograms and boxplots; compare farms by region, crop type, and season. Scatter plots and correlation can show associations between yield and environmental inputs.

B. Abnormal yields

Very low yield may reflect drought, pests, disease, or measurement error; very high yield may be genuine or misrecorded. Detect with IQR fences, robust z-scores, and region/crop-specific comparisons. Verify field records and weather. Keep valid extreme outcomes and investigate their causes rather than deleting them.

C. Organic versus chemical fertilizer

Let μO and μC be mean crop yields. H_0: $\mu O = \mu C$; H_1: $\mu O \neq \mu C$. Use an independent two-sample t-test or regression adjusted for soil, rainfall, crop, and farm characteristics. If farms are paired or matched, use a paired analysis. Report mean difference, CI, p-value, and note that observational differences may reflect farm selection.

Conclusion

Use the descriptive summaries to understand typical values and variation, investigate unusual observations before deciding how to treat them, and use a correctly specified hypothesis test with a confidence interval to support decisions. Interpret statistical results alongside practical context and data quality.

CASE STUDY 13: Employee Productivity Analysis

Scenario

Employee data include working hours, tasks completed, deadlines, experience, and performance rating.

A. Descriptive statistics

Summarize tasks completed and working hours using mean, median, SD, quartiles, and range. Ratings can be summarized with frequency tables and medians if ordinal. Use boxplots by experience level and scatter plots of hours versus tasks. Consider task complexity and quality, not only quantity.

B. Unusual task counts

Very high counts may reflect easier tasks, exceptional output, or duplicate logging; low counts may reflect leave, complex assignments, or system errors. Detect using IQR fences within comparable job roles and inspect task logs. Verify records and retain legitimate extremes. Compare productivity with quality and hours to avoid misleading conclusions.

C. Experienced versus new employees

Let μE and μN be mean tasks completed. H_0: $\mu E = \mu N$; H_1: $\mu E > \mu N$. Compare groups with Welch's t-test or regression adjusting for role, hours, and task complexity. Report the mean difference and CI. Do not infer experience causes greater productivity without a design that controls for these factors.

Conclusion

Use the descriptive summaries to understand typical values and variation, investigate unusual observations before deciding how to treat them, and use a correctly specified hypothesis test with a confidence interval to support decisions. Interpret statistical results alongside practical context and data quality.

CASE STUDY 14: Automobile Fuel Efficiency Analysis

Scenario

Vehicle data include fuel consumption, engine capacity, weight, mileage, and maintenance cost.

A. Descriptive statistics

Summarize mileage and fuel consumption with mean, median, SD, quartiles, and range. Plot histograms and boxplots by vehicle type and engine size. Scatter plots can show how mileage relates to weight, engine capacity, and maintenance cost.

B. Abnormal mileage

Extremely high mileage may reflect test conditions or recording/unit errors; low mileage may be caused by vehicle faults or driving conditions. Detect with boxplots, IQR fences, and checks against vehicle specifications. Confirm units and test protocols, correct verified errors, and retain valid unusual vehicles for investigation.

C. Hybrid versus petrol mileage

Let μH and μP be mean mileage. H_0: $\mu H = \mu P$; H_1: $\mu H > \mu P$. Compare hybrid and petrol vehicles with Welch's t-test or regression adjusted for weight, engine size, and model year. Report the mean difference, confidence interval, and p-value. A significant result supports higher average mileage in the sample but does not isolate the effect of powertrain unless comparisons are suitably controlled.

Conclusion

Use the descriptive summaries to understand typical values and variation, investigate unusual observations before deciding how to treat them, and use a correctly specified hypothesis test with a confidence interval to support decisions. Interpret statistical results alongside practical context and data quality.

CASE STUDY 15: Social Media Engagement Analysis

Scenario

Data include likes, shares, comments, followers, posting time, and engagement rate.

A. Descriptive statistics

Calculate mean, median, SD, quartiles, and range for likes, shares, comments, and engagement rate. Engagement counts are often strongly right-skewed, so median and IQR are useful. Compare rates rather than raw counts when audiences differ in size, and use histograms or log-scale plots.

B. Viral post outliers

Viral posts can pull the mean upward and dominate apparent engagement. Detect using boxplots, IQR fences, robust z-scores, and log-scale plots. Check for bots, paid promotion, and platform anomalies. Retain genuine viral posts, but report results with and without them or segment viral and ordinary posts.

C. Video versus image engagement

Let μV and μI be mean engagement rate for video and image posts. H_0: $\mu V = \mu I$; H_1: $\mu V > \mu I$. Compare posts using Welch's t-test or regression controlling for follower count, posting time, topic, and promotion. If engagement is count-based, choose a suitable count/rate model. Report effect size, CI, and p-value; avoid treating raw likes as directly comparable across audience sizes.

Conclusion

Use the descriptive summaries to understand typical values and variation, investigate unusual observations before deciding how to treat them, and use a correctly specified hypothesis test with a confidence interval to support decisions. Interpret statistical results alongside practical context and data quality.

CASE STUDY 16: Airline Passenger Satisfaction Analysis

Scenario

Passenger data include ticket price, flight delay, baggage waiting, seat comfort, and overall satisfaction.

A. Descriptive statistics

Summarize delay time, baggage wait, and satisfaction with mean, median, SD, quartiles, and range. Satisfaction ratings may be ordinal, so include frequency percentages and median. Use boxplots by route, cabin, and on-time/delayed status.

B. Long delays

Long delays can reduce average satisfaction and create a right-skewed delay distribution. Detect with IQR fences, boxplots, and time-series checks; verify flight logs and units. Retain genuine long delays, since they are important service failures. Report median and upper percentiles as well as the mean.

C. On-time versus delayed flights

Let μO and μD be mean satisfaction. H_0: $\mu O = \mu D$; H_1: $\mu O > \mu D$. Compare groups with Welch's t-test, or use ordinal regression if the rating scale is ordinal. Adjust for route, cabin, and passenger type where possible. Report difference, CI, and p-value; a significant result indicates an association between punctuality and satisfaction.

Conclusion

Use the descriptive summaries to understand typical values and variation, investigate unusual observations before deciding how to treat them, and use a correctly specified hypothesis test with a confidence interval to support decisions. Interpret statistical results alongside practical context and data quality.

CASE STUDY 17: Smart City Traffic Analysis

Scenario

Road-level data include vehicle count, average speed, signal waiting time, accident count, and pollution.

A. Descriptive statistics

Calculate mean, median, SD, quartiles, and range for vehicle count, waiting time, and speed. Use histograms, boxplots, and time-series plots by road and hour. Scatter plots can explore relationships among traffic volume, speed, accidents, and pollution.

B. Abnormal traffic values

Traffic spikes during accidents, festivals, or roadblocks may be real events, not errors. Detect unusual values with IQR fences, robust z-scores, and time-series residuals relative to road and time-of-day baselines. Verify sensor records and incident reports. Retain real events and label them so they can inform emergency planning.

C. Peak versus non-peak waiting time

Let μP and μN be mean signal waiting time. H_0: $\mu P = \mu N$; H_1: $\mu P > \mu N$. Use Welch's t-test or paired route/time comparisons. If observations repeat by road, use a model that accounts for clustering. Report difference, CI, and p-value, then prioritize signal timing and enforcement where delays are greatest.

Conclusion

Use the descriptive summaries to understand typical values and variation, investigate unusual observations before deciding how to treat them, and use a correctly specified hypothesis test with a confidence interval to support decisions. Interpret statistical results alongside practical context and data quality.

CASE STUDY 18: Hotel Booking and Customer Review Analysis

Scenario

Hotel data include room price, length of stay, customer rating, cancellation status, and service feedback.

A. Descriptive statistics

Summarize room price, stay length, and ratings with mean, median, SD, quartiles, and range. Prices are often skewed, so use median and IQR. Compare ratings by hotel, room type, season, and booking status using boxplots and frequency tables.

B. Outliers

Very high room prices may occur during events or premium room bookings; very low ratings may reflect genuine poor service or mistaken entries. Detect with IQR fences and boxplots within comparable room/season groups. Verify booking and review records. Retain valid extreme values and analyse event periods separately when estimating typical prices.

C. Stay length and ratings

Let μL be mean rating for stays longer than three days and μS for stays of one or two days. H_0: $\mu L = \mu S$; H_1: $\mu L \neq \mu S$. Compare groups with Welch's t-test or an ordinal/non-parametric test suitable for ratings. Report difference, CI, and p-value. Account for hotel, price, and customer type; the test checks association, not necessarily causation.

Conclusion

Use the descriptive summaries to understand typical values and variation, investigate unusual observations before deciding how to treat them, and use a correctly specified hypothesis test with a confidence interval to support decisions. Interpret statistical results alongside practical context and data quality.

CASE STUDY 19: Fitness App User Health Analysis

Scenario

Fitness data include steps, calories burned, workout duration, heart rate, sleep hours, and goal completion.

A. Descriptive statistics

Calculate mean, median, SD, quartiles, and range for steps, calories, workout duration, and sleep. Use daily/weekly time-series charts and histograms. Summarize by user group while accounting for repeated measurements from the same person.

B. Abnormal activity and heart rate

Extremely long workouts may reflect tracking left on, and abnormal heart-rate values may be sensor errors or genuine health events. Detect using physiologically informed checks, IQR fences, and time-series consistency. Compare with raw sensor/device records. Do not automatically discard unusual heart rates; flag potentially important readings and follow appropriate safety procedures.

C. Sleep at least seven hours versus less

Let $\mu 7$ be mean calories burned for users sleeping ≥ 7 hours and μL for users sleeping <7 hours. H_0: $\mu 7 = \mu L$; H_1: $\mu 7 > \mu L$. Compare groups with Welch's t-test or a regression accounting for age, activity, and repeated observations. Report effect size, CI, and p-value. The result is observational and cannot prove that longer sleep causes more calories burned.

Conclusion

Use the descriptive summaries to understand typical values and variation, investigate unusual observations before deciding how to treat them, and use a correctly specified hypothesis test with a confidence interval to support decisions. Interpret statistical results alongside practical context and data quality.

CASE STUDY 20: Online Examination Performance Analysis

Scenario

Exam data include score, completion time, number of attempts, internet interruptions, and feedback.

A. Descriptive statistics

Summarize exam scores and completion time with mean, median, SD, quartiles, and range. Summarize attempts with counts, median, and frequency table. Use histograms and boxplots; compare results by interruption count and attempt number.

B. Unusual completion time and attempts

Extremely short completion time may indicate guessing, prior knowledge, or a timer/logging issue; many attempts may indicate technical problems or repeated retries. Detect using IQR fences, robust z-scores, and checks against attempt and interruption logs. Investigate before excluding records, and distinguish academic behaviour from system failures.

C. Fewer versus frequent interruptions

Let μF be mean score for students with fewer interruptions and μH for students with frequent interruptions. H_0: $\mu F = \mu H$; H_1: $\mu F > \mu H$. Define the interruption groups in advance, compare with Welch's t-test, or use regression with score as outcome and interruption count as predictor. Report mean difference, CI, and p-value. Account for prior ability, device, and exam conditions; association alone does not prove interruptions caused lower scores.

Conclusion

Use the descriptive summaries to understand typical values and variation, investigate unusual observations before deciding how to treat them, and use a correctly specified hypothesis test with a confidence interval to support decisions. Interpret statistical results alongside practical context and data quality.